Brief Technical Report

WasteSort: supervised waste-material image classification

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1. Objective and dataset readiness

The objective is to classify a single item photograph into cardboard, glass, metal, paper, plastic or trash and test whether a small Python prototype is feasible. The initial study uses secondary TrashNet data. No primary campus observations have been collected; a primary dataset is therefore not submitted or claimed. If primary collection is required for this milestone, it remains outstanding.

Gary Thung and Mindy Yang collected the source images using iPhone cameras, placing individual items on white posterboard under sunlight and/or room lighting. Their resized archive contains 2,527 RGB JPEG images at 512 x 384 pixels [1]. We retrieved the pinned archive on October 20, 2026 [2] and recorded its SHA-256. The supplied source archive and inventory preserve provenance.

The inventory CSV contains all 2,527 source records; the frozen-split CSV contains 2,521 retained observations. The feature CSV contains the same 2,521 observations, five metadata/target columns and 1,764 HOG features computed from verified image bytes. Features are numerical edge-orientation descriptors, not invented survey observations. The neural models consume RGB image tensors rather than the HOG CSV. The target is a six-category material label, encoded 0-5 in the documented class order.

2. Cleaning and preprocessing

Missing data: all source images decoded and had known folder labels; no pixel or target imputation was needed. Blank exclusion reasons on included records and blank split/group fields on excluded records mean not applicable. Duplicates: six files in three identical-pixel pairs had conflicting labels; all six were excluded. Forty-five perceptual-hash candidate pairs were grouped conservatively within partitions and received later AI-assisted visual notes. This does not establish a complete independent human audit.

Outliers: no statistical outlier threshold or deletion was applied. Unusual item appearances were retained unless covered by the documented conflicting-label rule. No learned feature-selection stage was used. HOG models use 64 x 64 grayscale inputs and 1,764 descriptors; SVM and KNN fit StandardScaler on training rows only. Random Forest and Decision Tree use unscaled descriptors.

Neural inputs are resized to 160 x 160 RGB. CNN inputs are in [0,1]; MobileNetV2 additionally uses mean [0.485,0.456,0.406] and standard deviation [0.229,0.224,0.225]. Training-only augmentation uses horizontal flips, rotations and brightness/contrast changes. Validation and test transformations are deterministic. Square resizing changes aspect ratio and is a baseline design choice.

3. Partition and leakage control

StratifiedGroupKFold with 20 folds and seed 42 assigns folds 0-13 to training, 14-16 to validation and 17-19 to testing: 1,774 / 371 / 376 images. The approximately 70/15/15 proportions preserve candidate groups rather than forcing exact percentages. Paths, group IDs, labels and split fields must not become predictor features. The test partition was evaluated after deployment selection; it is now exposed and must not be treated as unseen during follow-up tuning.

Model, findings and recommendations

4. Initial implementation and actual training

The primary comparison trained HOG + SVM, HOG + Random Forest, a small CNN and MobileNetV2 in Python. MobileNetV2 is appropriate because pretrained visual features can be adapted to a relatively small labeled image dataset. Its ImageNet backbone, including batch normalization, was frozen; a dropout layer and six-class linear head were trained. The optimizer used learning rate 0.001 and weight decay 0.0001, batch size 32, a 20-epoch budget and patience 5. The best selected epoch was 8. PyTorch/torchvision implement the neural models; Scikit-learn implements the classical models.

The small CNN uses three convolutional blocks and a fixed pooling operation equivalent to the planned 4 x 4 adaptive output for the fixed input dimensions. Validation macro F1 selected the checkpoint and deployment model. KNN and Decision Tree are supporting classification exercises; weighted CNN is a separate ablation. Linear Regression on the diabetes benchmark is a separate educational task, not WasteSort training data or clinical guidance.

5. Preliminary findings

MobileNetV2 achieved 84.57% test accuracy, 82.05% macro precision, 80.58% macro recall and 81.22% macro F1. This exceeded the other primary models in this recorded test comparison after validation-based selection. A working Streamlit interface loads the saved model, accepts a photo and displays the predicted material. The results support feasibility of image-based material classification under the recorded dataset conditions.

The strongest limitation is transfer to real campus scenes: TrashNet has controlled backgrounds and single items. Trash recall was 55.56% (10/18), with few test examples in that class. Other limitations include class imbalance, possible undetected duplicates, label uncertainty, composite materials, one split/seed, uncalibrated scores and no reliable rejection of unrelated objects. Descriptive group-bootstrap intervals do not establish formal hypothesis rejection. Cross-platform CNN differences in Colab further limit claims of exact numerical reproducibility.

6. Recommendations and next steps

Collect permissioned campus images with independent labels and a documented sampling plan. Evaluate them separately using evaluate_photos.py; do not merge those scores into this benchmark. Collect more diverse minority-class examples for a separately versioned training study, review labels and duplicates, and examine repeated splits/seeds before making stronger comparisons. Test uncertainty/rejection behavior using development data. Obtain official campus rules before displaying bin recommendations. Do not equate material prediction with verified disposal safety.

The implementation differs from the proposal in using PyTorch rather than TensorFlow/Keras, preserving duplicate groups in an approximate split, and reporting descriptive comparisons rather than formal hypothesis tests. These differences are documented for instructor review. Research-progress evidence does not imply instructor approval or completed campus validation.

7. Reproduction, ethics and sources

02_Source_Code includes the complete runnable project with trained checkpoints, the runnable full notebook and its actual executed Colab copy. 03_Results_Evidence preserves original metrics and predictions. Data are publicly sourced item images; future primary collection must use permissioned, non-sensitive photos. AI assisted implementation, execution and report preparation; independent human verification is not implied.

[1] Thung, G., and Yang, M. TrashNet dataset and project repository. https://github.com/garythung/trashnet (source documentation checked November 7, 2026).

[2] Creators' dataset archive, pinned revision 94cd17e: https://huggingface.co/datasets/garythung/trashnet/resolve/94cd17e/dataset-resized.zip . SHA-256 and retrieval date: 01_Datasets/Source_Provenance.json.

[3] Goal Diggers. WasteSort source, frozen protocol and experimental records. https://github.com/haerinfan/wastesort . Original run October 20, 2026; separate full Colab run October 26, 2026.